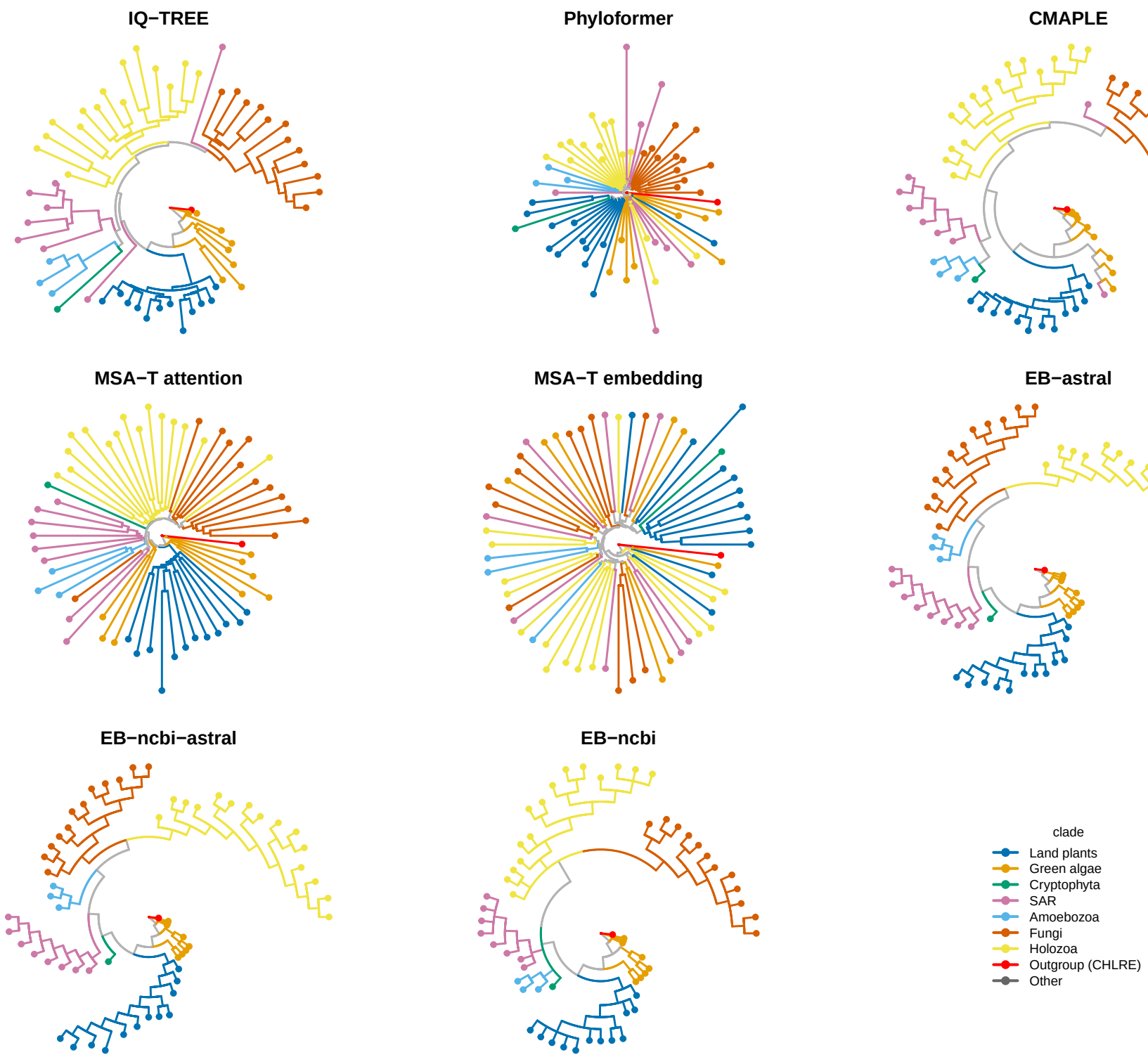
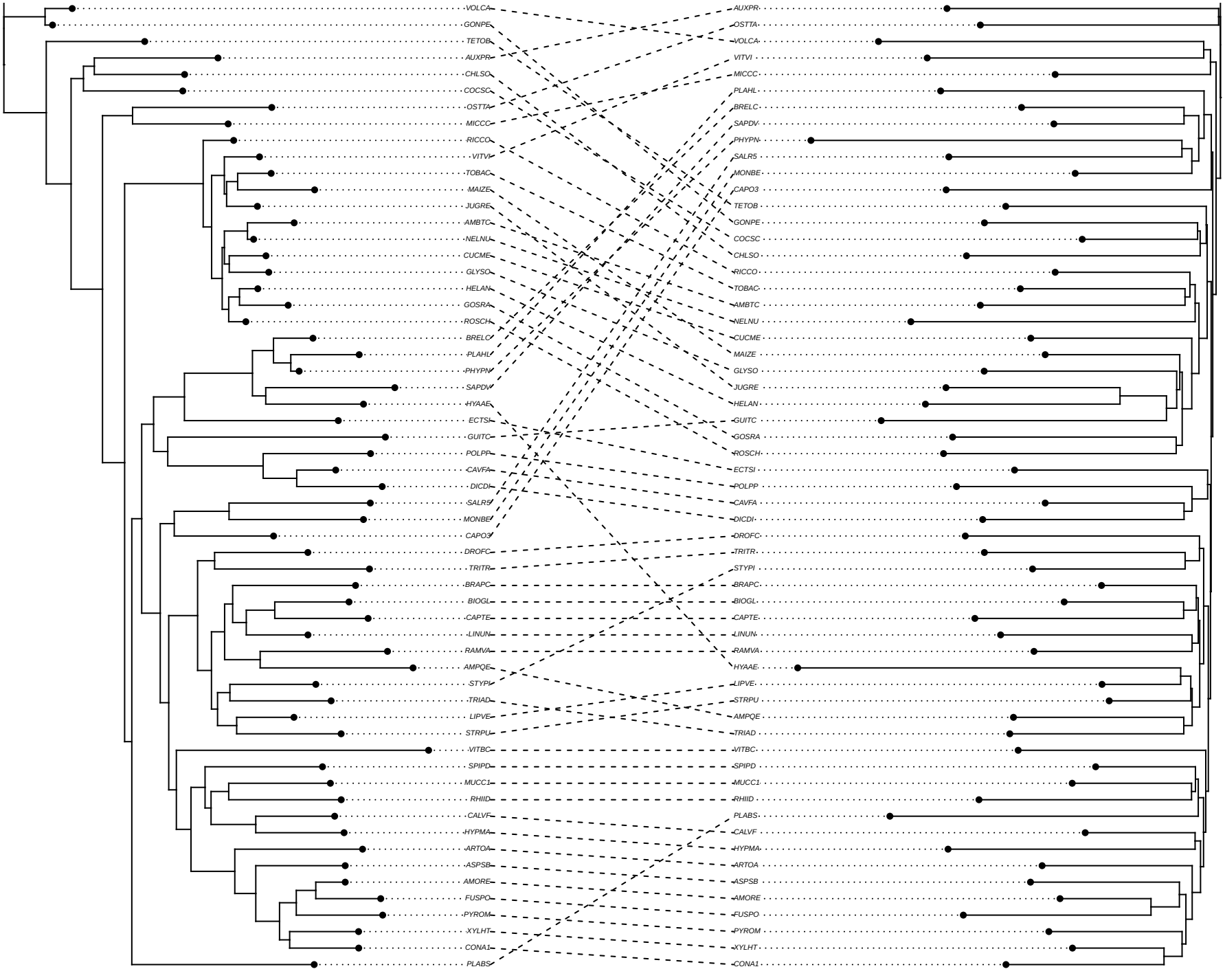
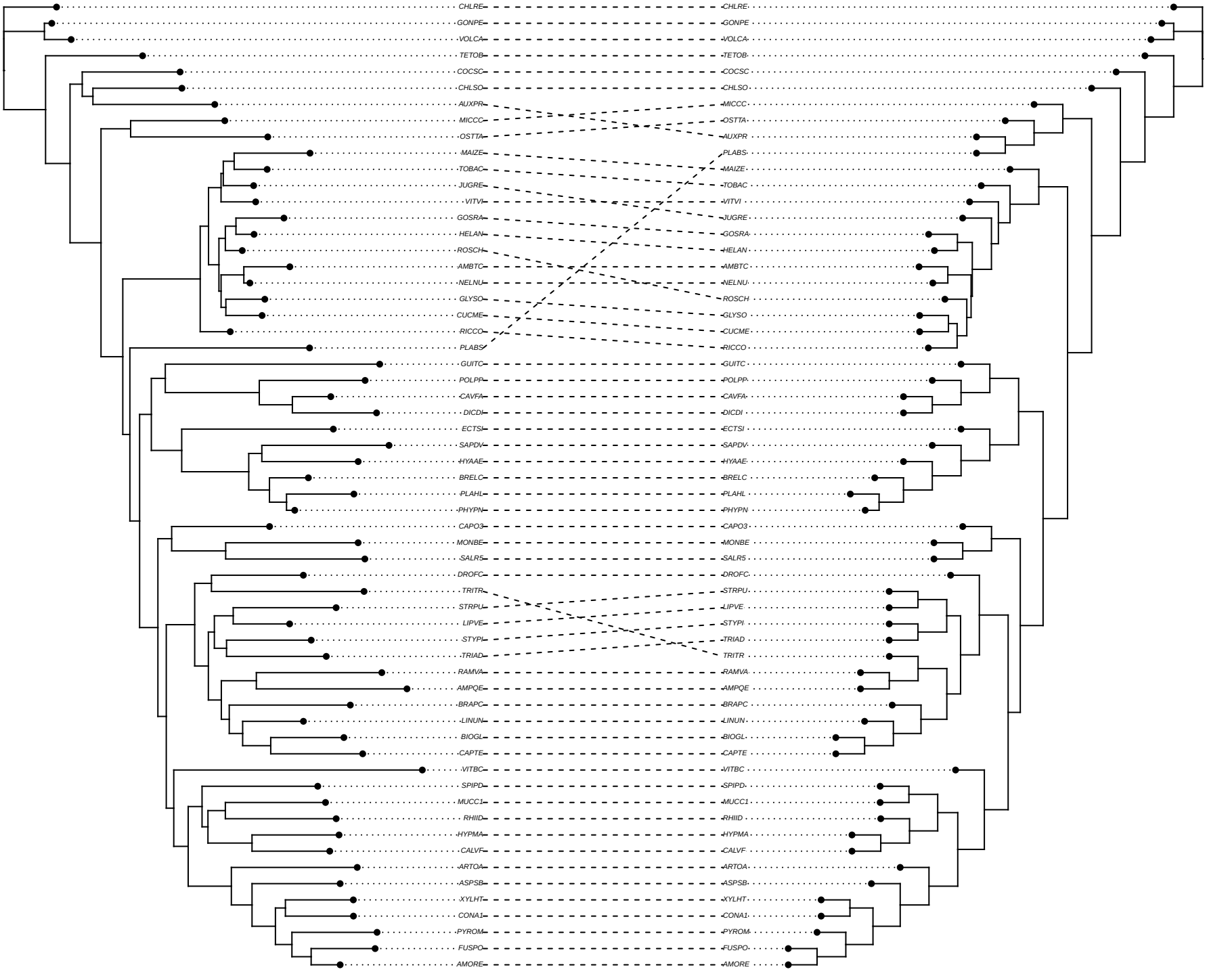
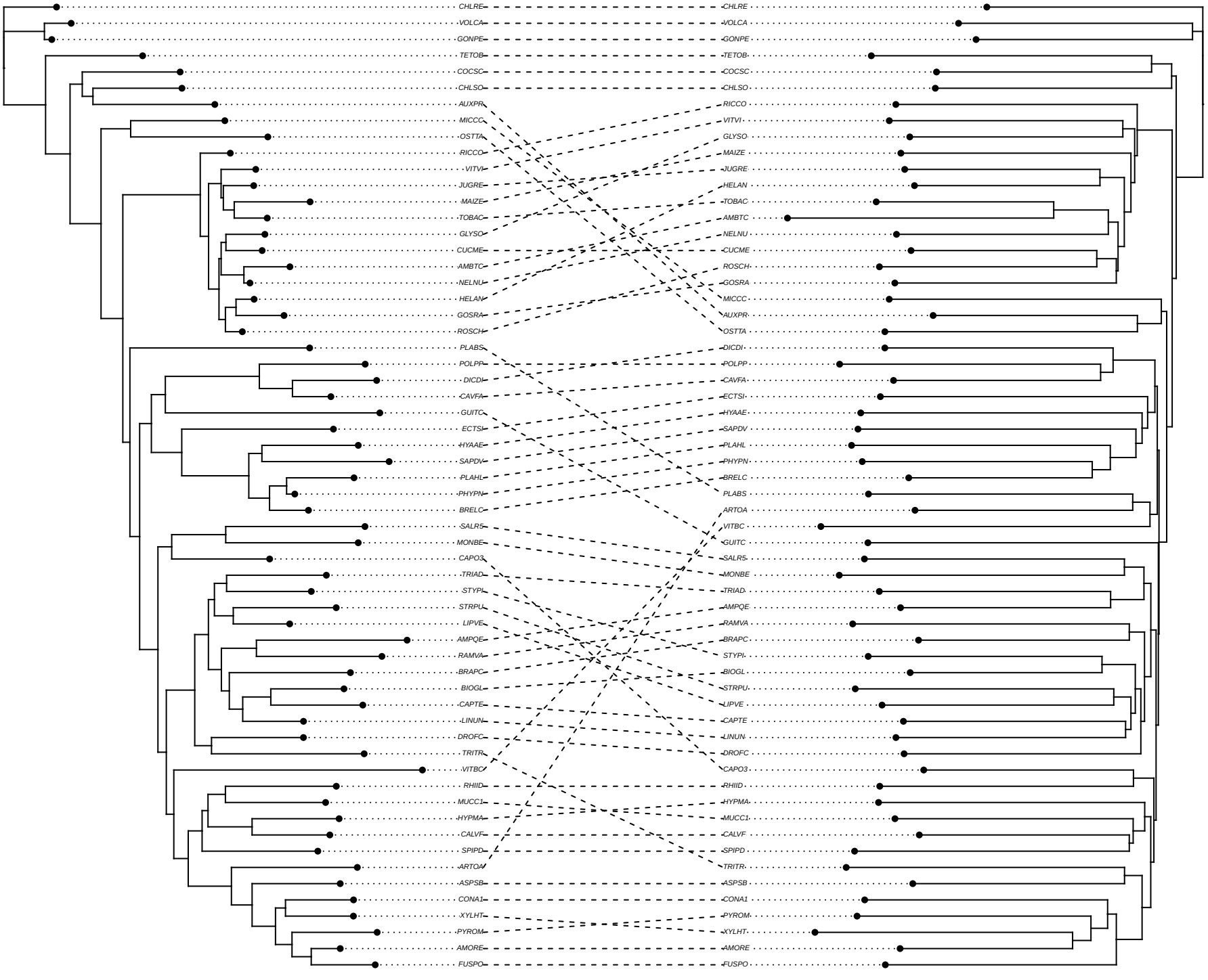
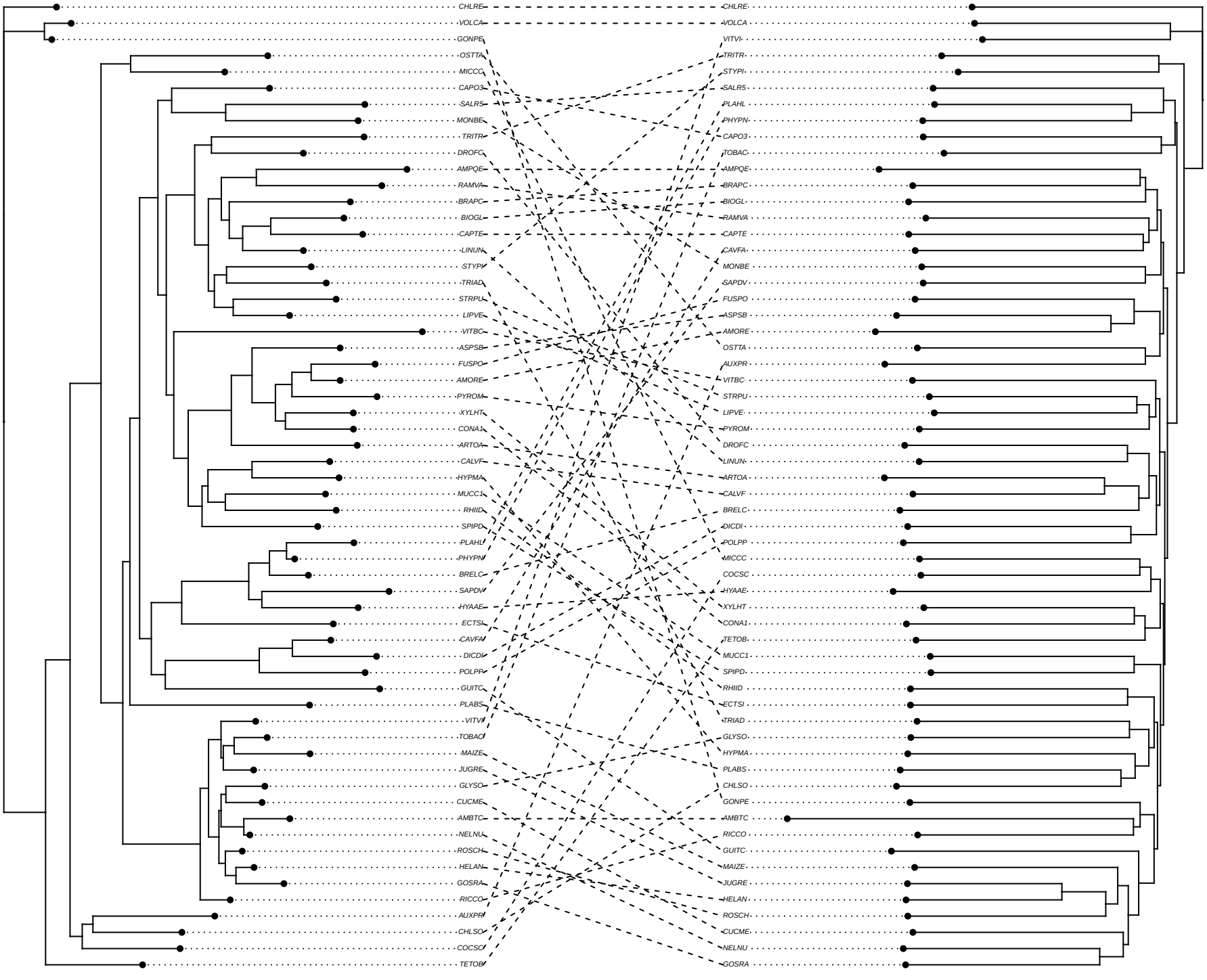

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
EB−astral
EB−ncbi−astral
EB−ncbi
clade
Land plants
Green algae
Cryptophyta
SAR
Amoebozoa
Fungi
Holozoa
Outgroup (CHLRE)
Other


IQ−TREE vs Phyloformer
VOLCA
GONPE
TETOB
AUXPR
CHLSO
COCSC
OSTTA
MICCC
RICCO
VITVI
TOBAC
MAIZE
JUGRE
AMBTC
NELNU
CUCME
GLYSO
HELAN
GOSRA
ROSCH
BRELC
PLAHL
PHYPN
SAPDV
HYAAE
ECTSI
GUITC
POLPP
CAVFA
DICDI
SALR5
MONBE
CAPO3
DROFC
TRITR
BRAPC
BIOGL
CAPTE
LINUN
RAMVA
AMPQE
STYPI
TRIAD
LIPVE
STRPU
VITBC
SPIPD
MUCC1
RHIID
CALVF
HYPMA
ARTOA
ASPSB
AMORE
FUSPO
PYROM
XYLHT
CONA1
PLABS
AUXPR
OSTTA
VOLCA
VITVI
MICCC
PLAHL
BRELC
SAPDV
PHYPN
SALR5
MONBE
CAPO3
TETOB
GONPE
COCSC
CHLSO
RICCO
TOBAC
AMBTC
NELNU
CUCME
MAIZE
GLYSO
JUGRE
HELAN
GUITC
GOSRA
ROSCH
ECTSI
POLPP
CAVFA
DICDI
DROFC
TRITR
STYPI
BRAPC
BIOGL
CAPTE
LINUN
RAMVA
HYAAE
LIPVE
STRPU
AMPQE
TRIAD
VITBC
SPIPD
MUCC1
RHIID
PLABS
CALVF
HYPMA
ARTOA
ASPSB
AMORE
FUSPO
PYROM
XYLHT
CONA1

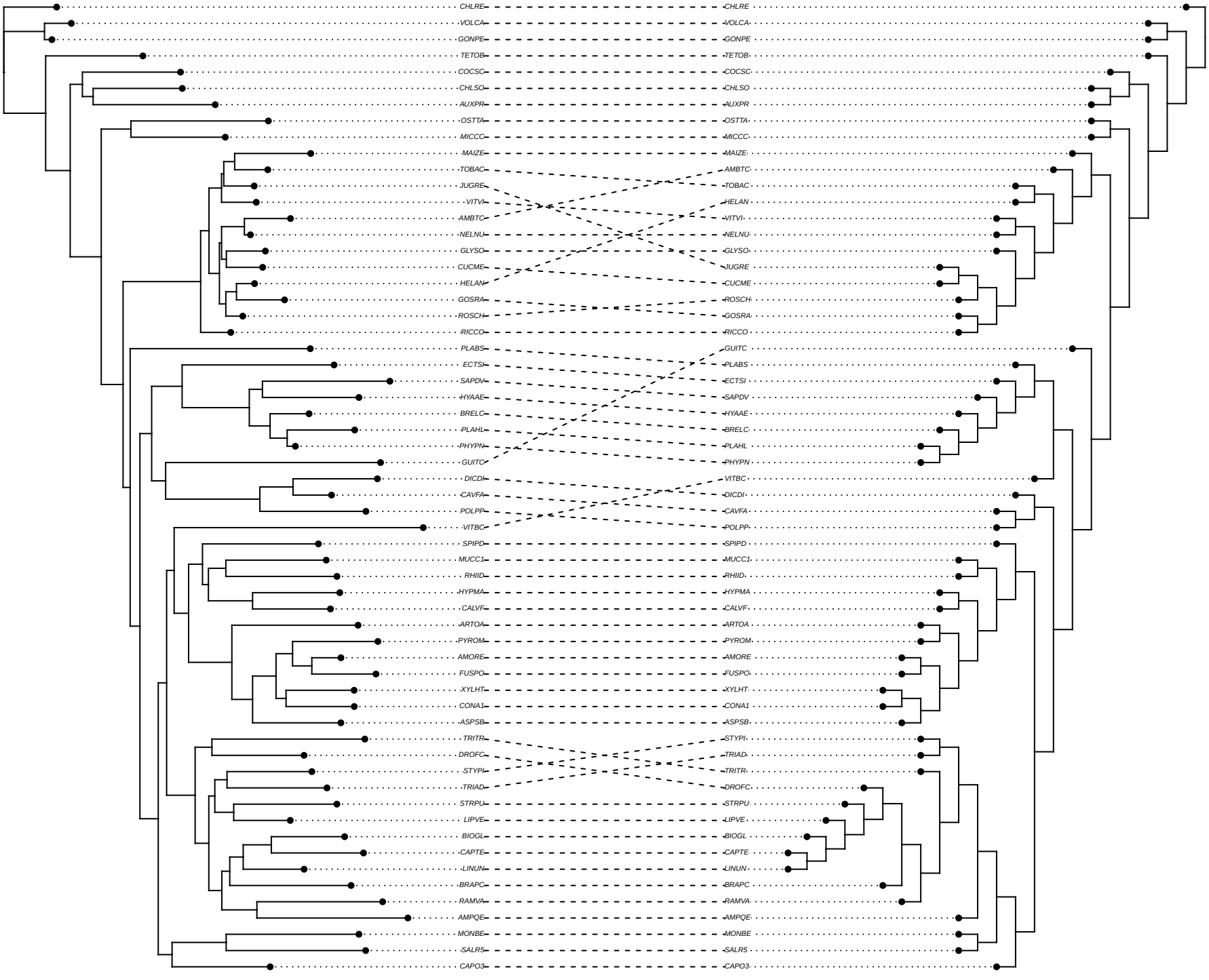
IQ-TREE vs CMAPLE



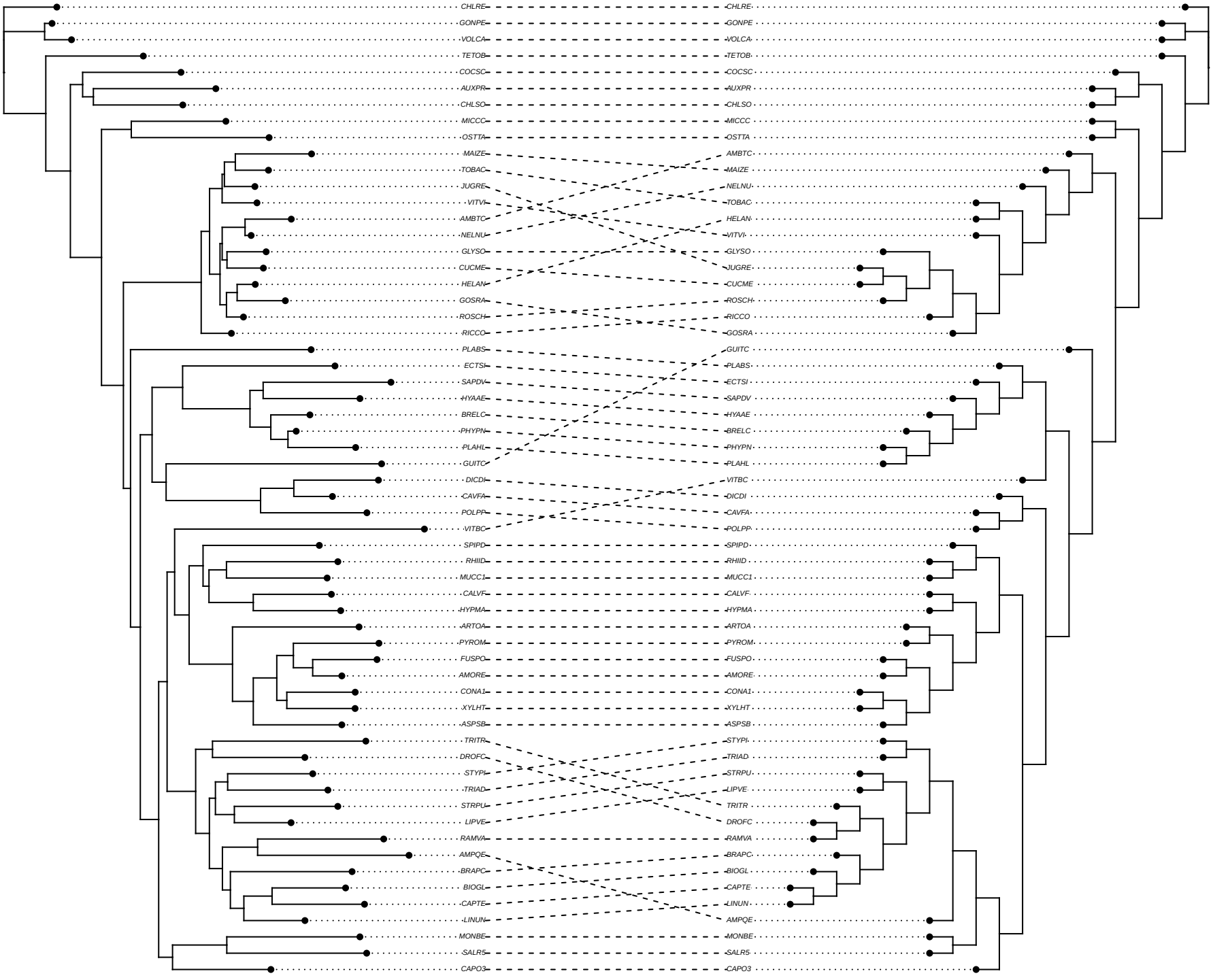

IQ−TREE vs MSA−Pattention
CHLRE
VOLCA
GONPE
TETOB
COCSC
CHLSO
AUXPR
MICCC
OSTTA
RICCO
VITVI
JUGRE
MAIZE
TOBAC
GLYSO
CUCME
AMBTC
NELNU
HELAN
GOSRA
ROSCH
PLABS
POLPP
DICDI
CAVFA
GUITC
ECTSI
HYAAE
SAPDV
PLAHL
PHYPN
BRELC
SALR5
MONBE
CAPO3
TRIAD
STYPI
STRPU
LIPVE
AMPQE
RAMVA
BRAPC
BIOGL
CAPTE
LINUN
DROFC
TRITR
VITBC
RHIID
MUCC1
HYPMA
CALVF
SPIPD
ARTOA
ASPSB
CONA1
XYLHT
PYROM
AMORE
FUSPO


IQ−TREE vs MSA−T embedding
CHLRE
VOLCA
GONPE
OSTTA
MICCC
CAPO3
SALR5
MONBE
TRITR
DROFC
AMPQE
RAMVA
BRAPC
BIOGL
CAPTE
LINUN
STYPI
TRIAD
STRPU
LIPVE
VITBC
ASPSB
FUSPO
AMORE
PYROM
XYLHT
CONA1
ARTOA
CALVF
HYPMA
MUCC1
RHIID
SPIPD
PLAHL
PHYPN
BRELC
SAPDV
HYAAE
ECTSI
CAVFA
DICDI
POLPP
GUITC
PLABS
VITVI
TOBAC
MAIZE
JUGRE
GLYSO
CUCME
AMBTC
NELNU
ROSCH
HELAN
GOSRA
RICCO
AUXPR
CHLSO
COCSC
TETOB
CHLRE
VOLCA
VITVI
TRITR
STYPI
SALR5
PLAHL
PHYPN
CAPO3
TOBAC
AMPQE
BRAPC
BIOGL
RAMVA
CAPTE
CAVFA
MONBE
SAPDV
FUSPO
ASPSB
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PYROM
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BRELC
DICDI
POLPP
MICCC
COCSC
HYAAE
XYLHT
CONA1
TETOB
MUCC1
SPIPD
RHIID
ECTSI
TRIAD
GLYSO
HYPMA
PLABS
CHLSO
GONPE
AMBTC
RICCO
GUITC
MAIZE
JUGRE
HELAN
ROSCH
CUCME
NELNU
GOSRA

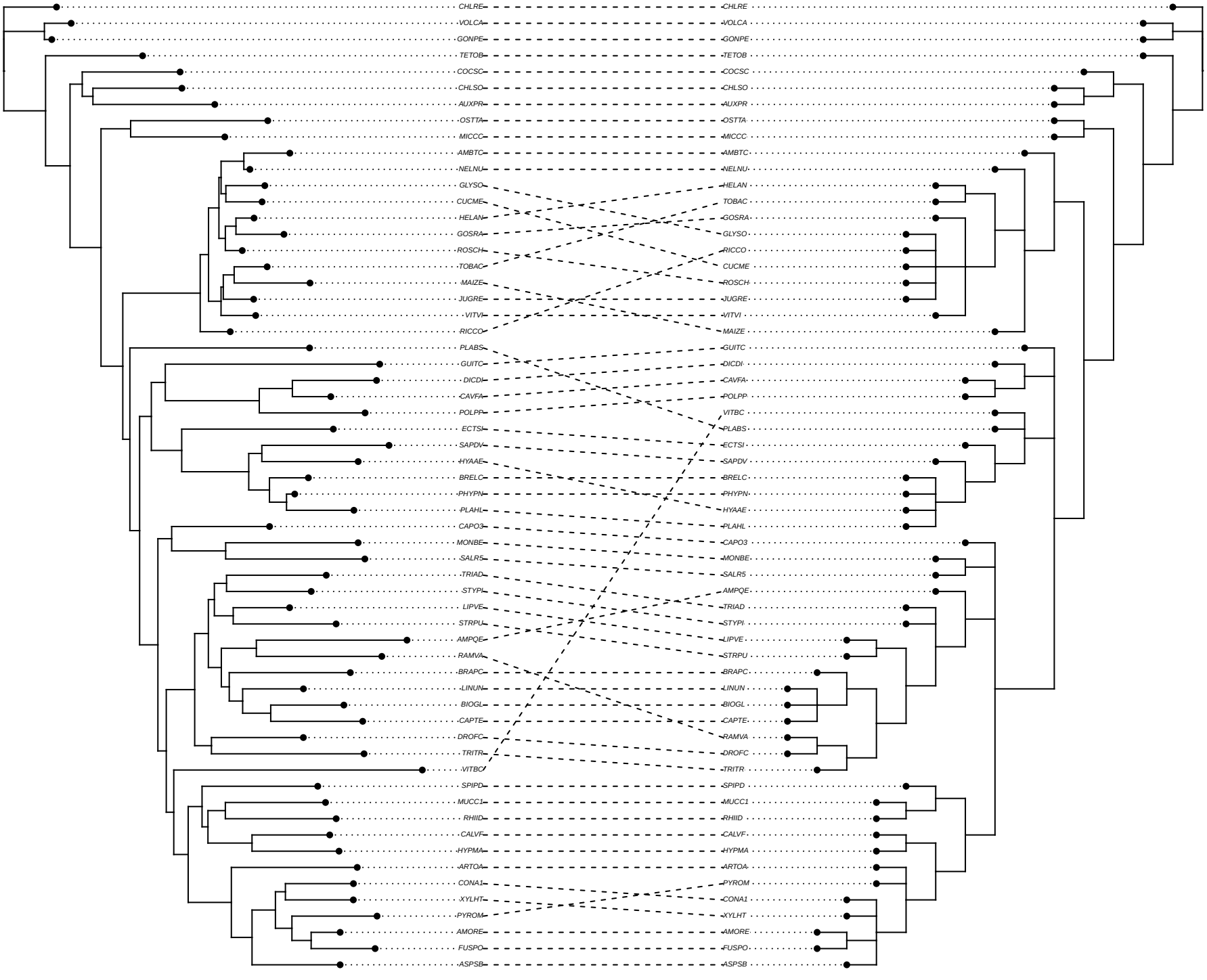
IQ−TREE vs EB−astral

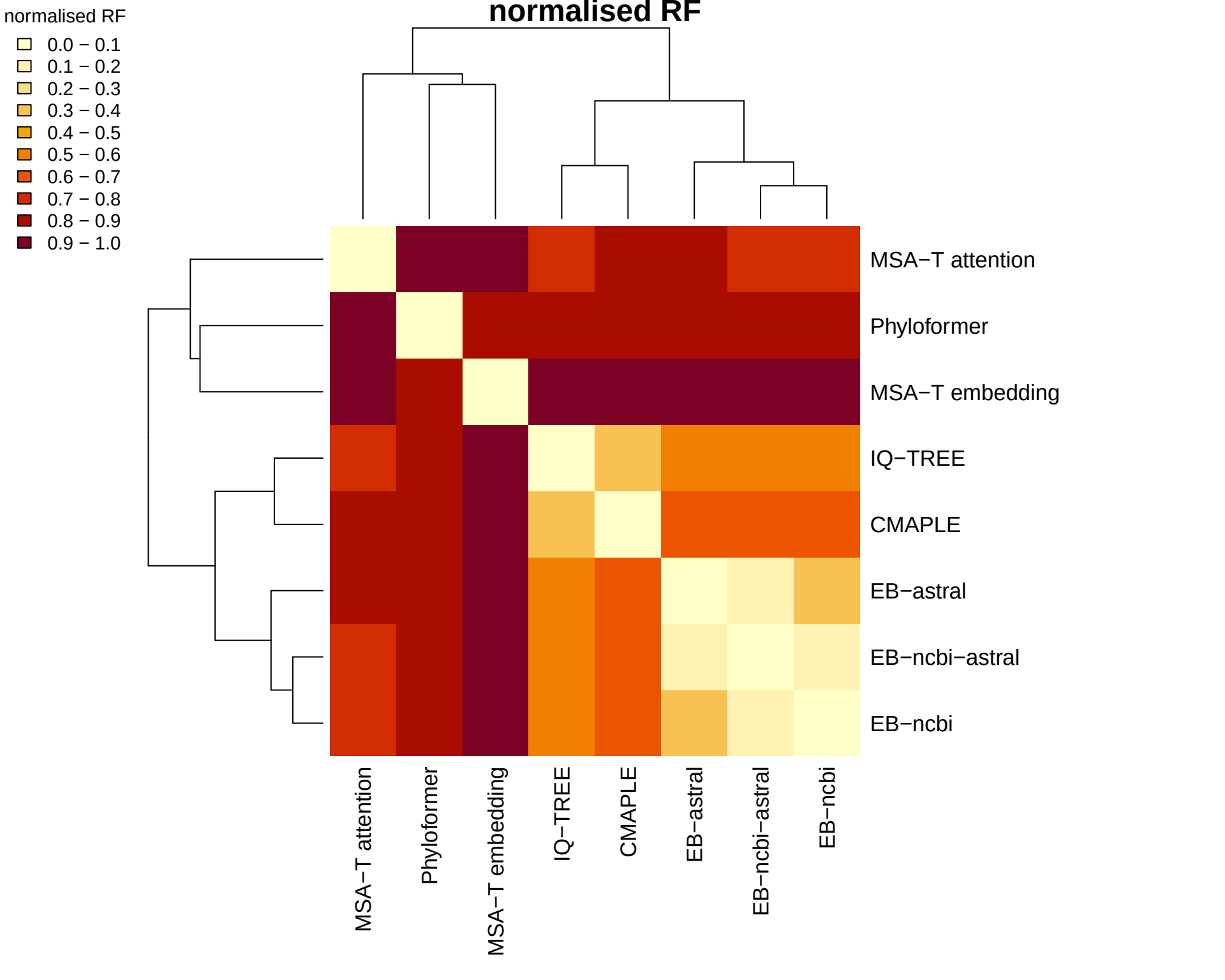


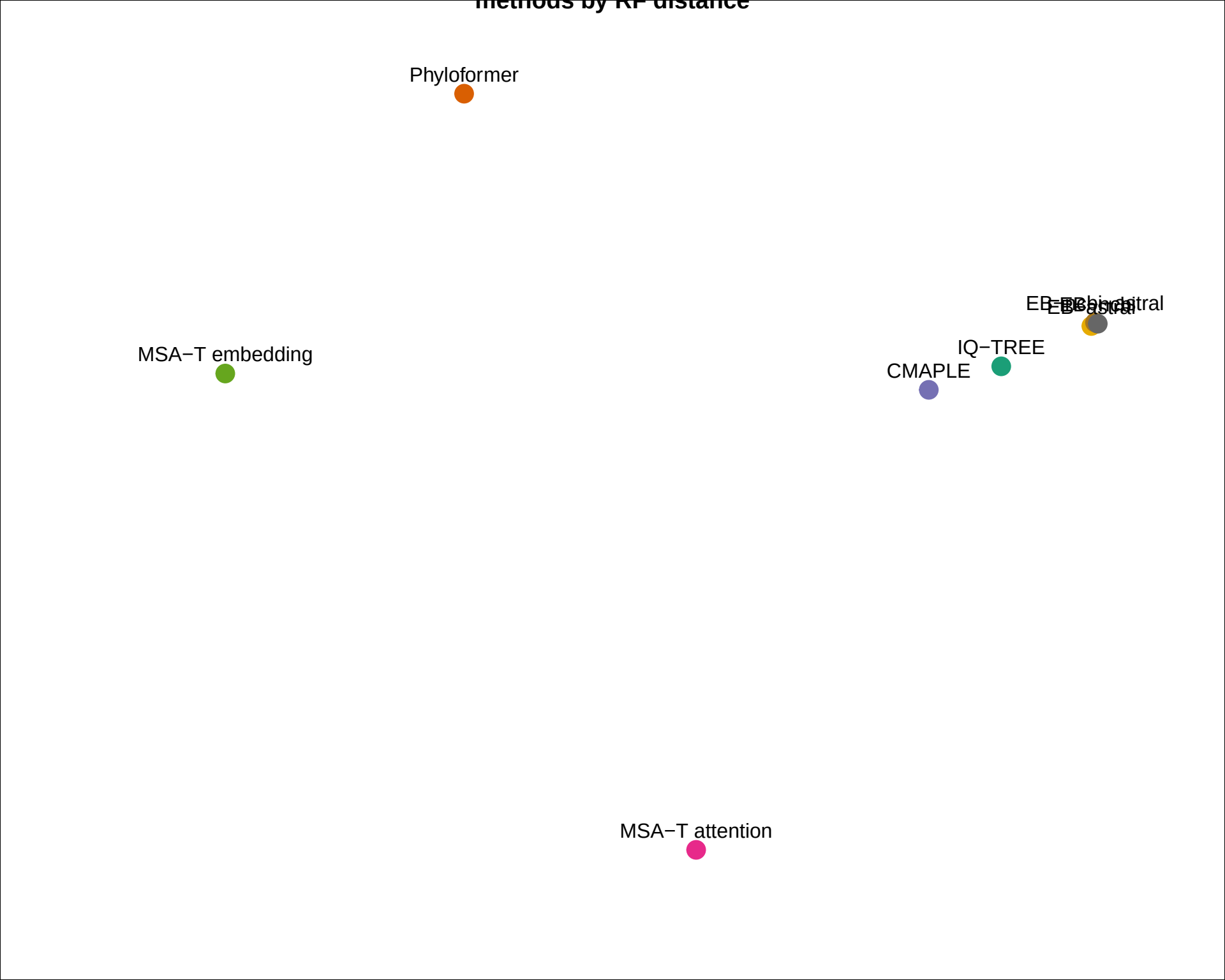
IQ-TREE vs EB-ncbi-astral

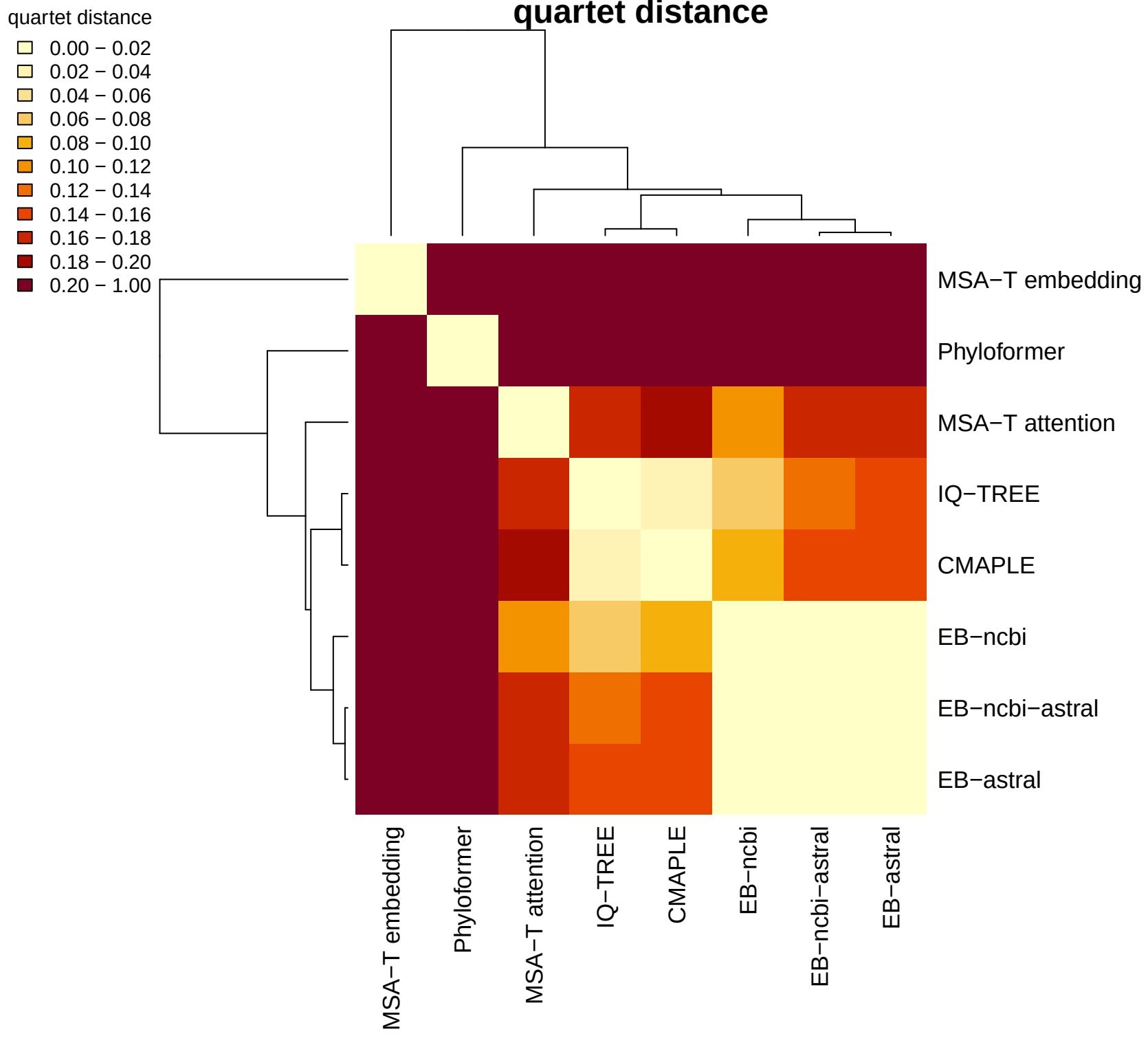


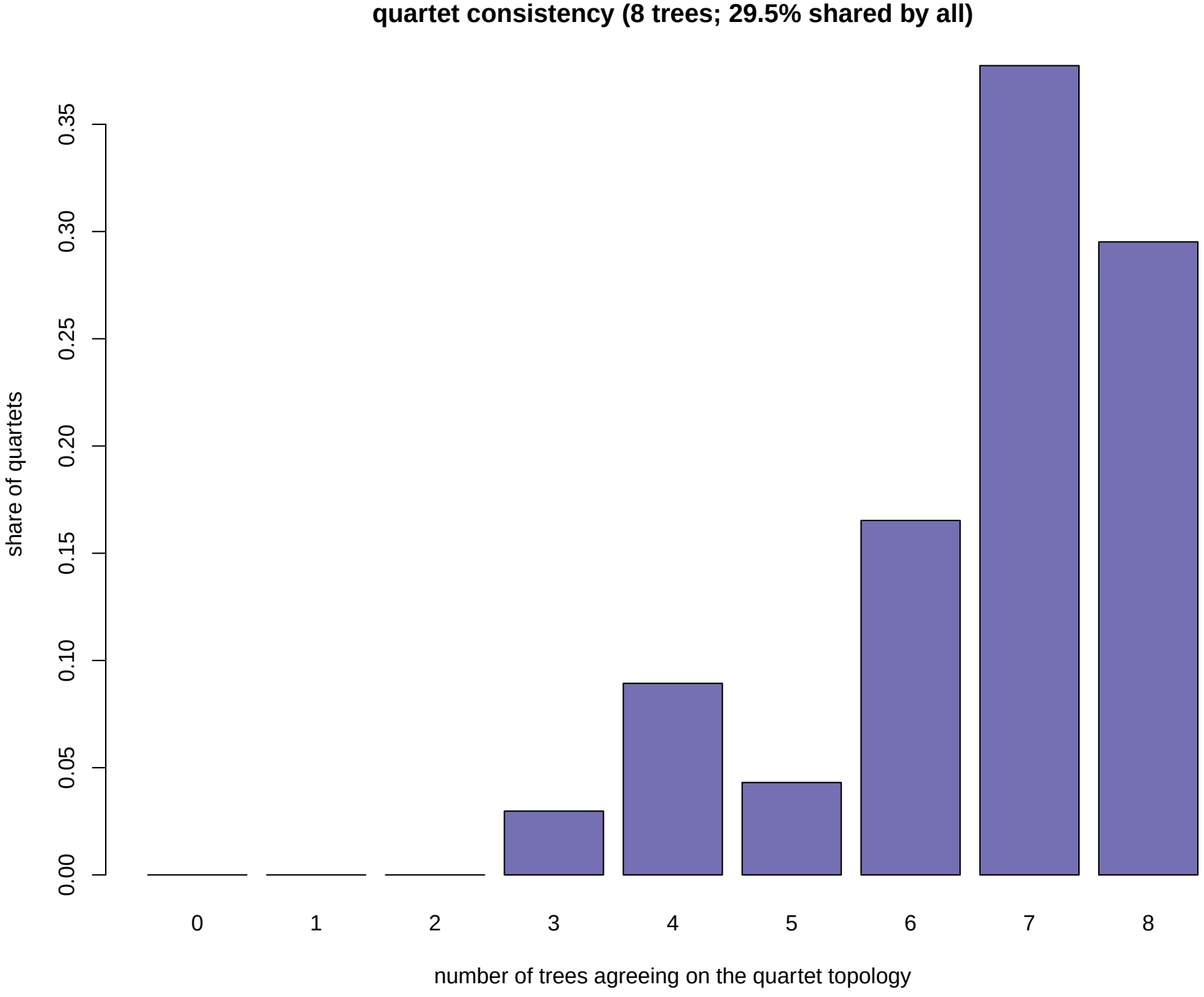
IQ-TREE vs EB-ncbi

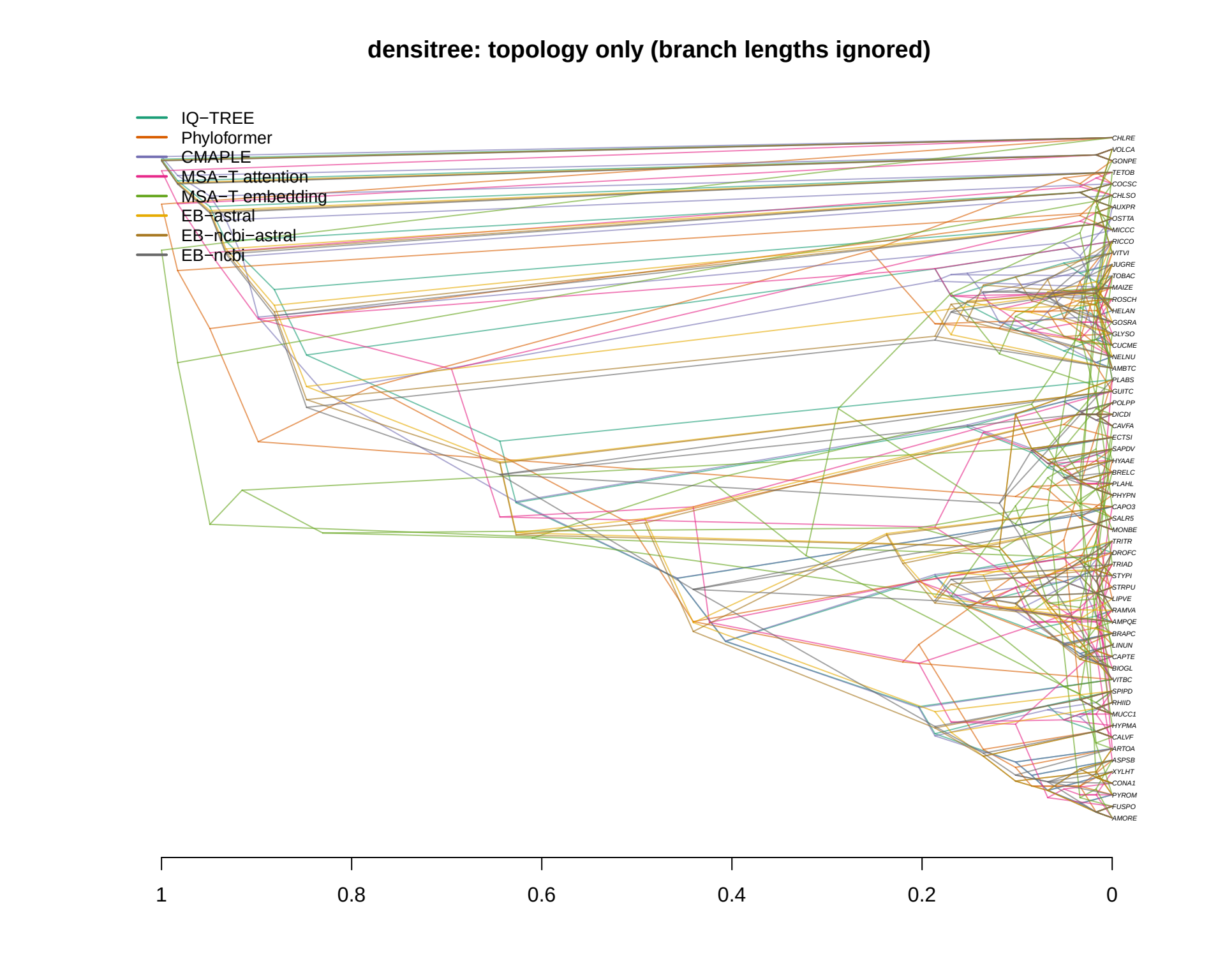


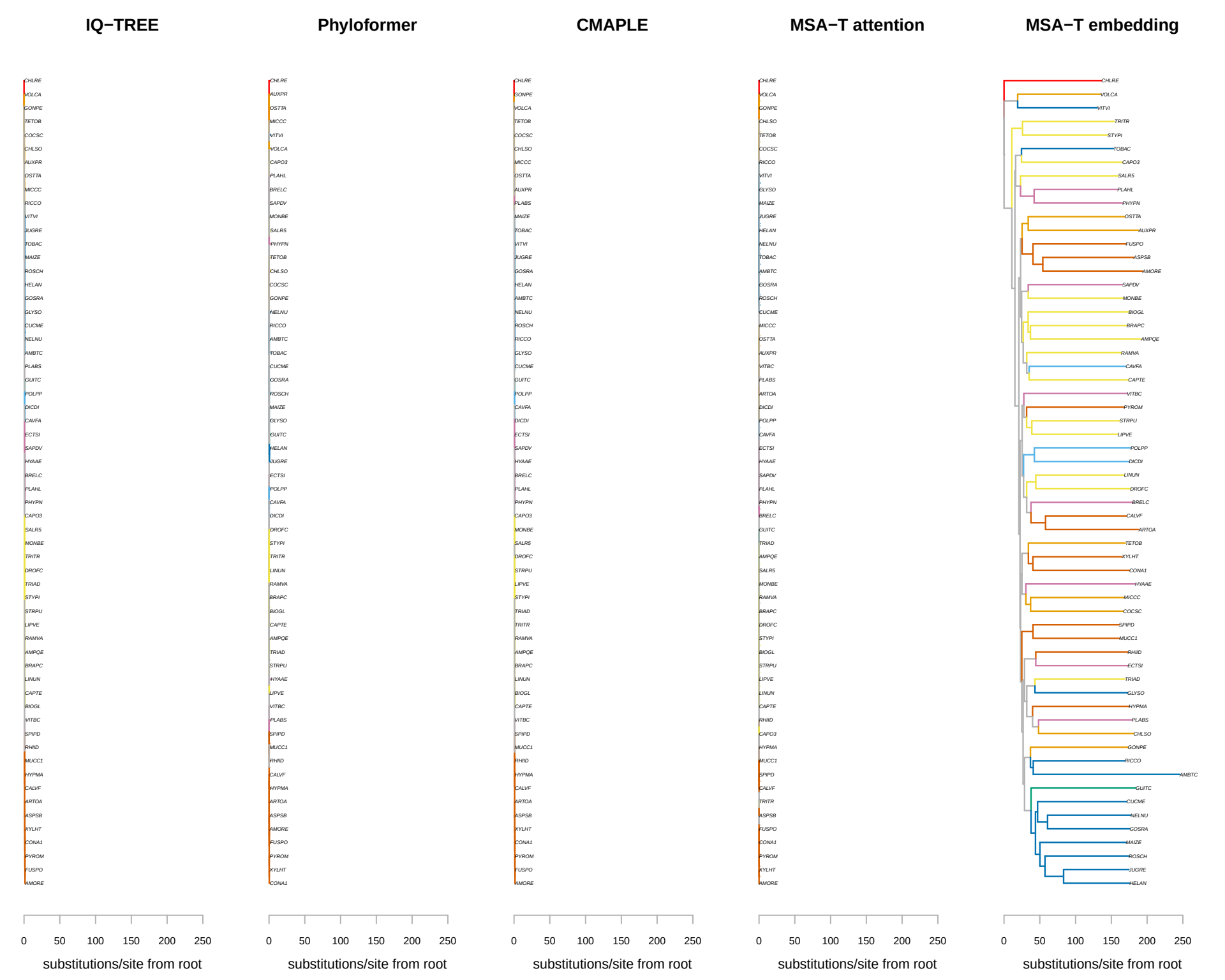

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T attention
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
EB−astral
EB−ncbi−astral
EB−ncbi
MSA−T attention
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
EB−astral
EB−ncbi−astral
EB−ncbi


methods by RF distance
Phyloformer
MSA−T embedding
IQ−TREE
CMAPLE
MSA−T attention

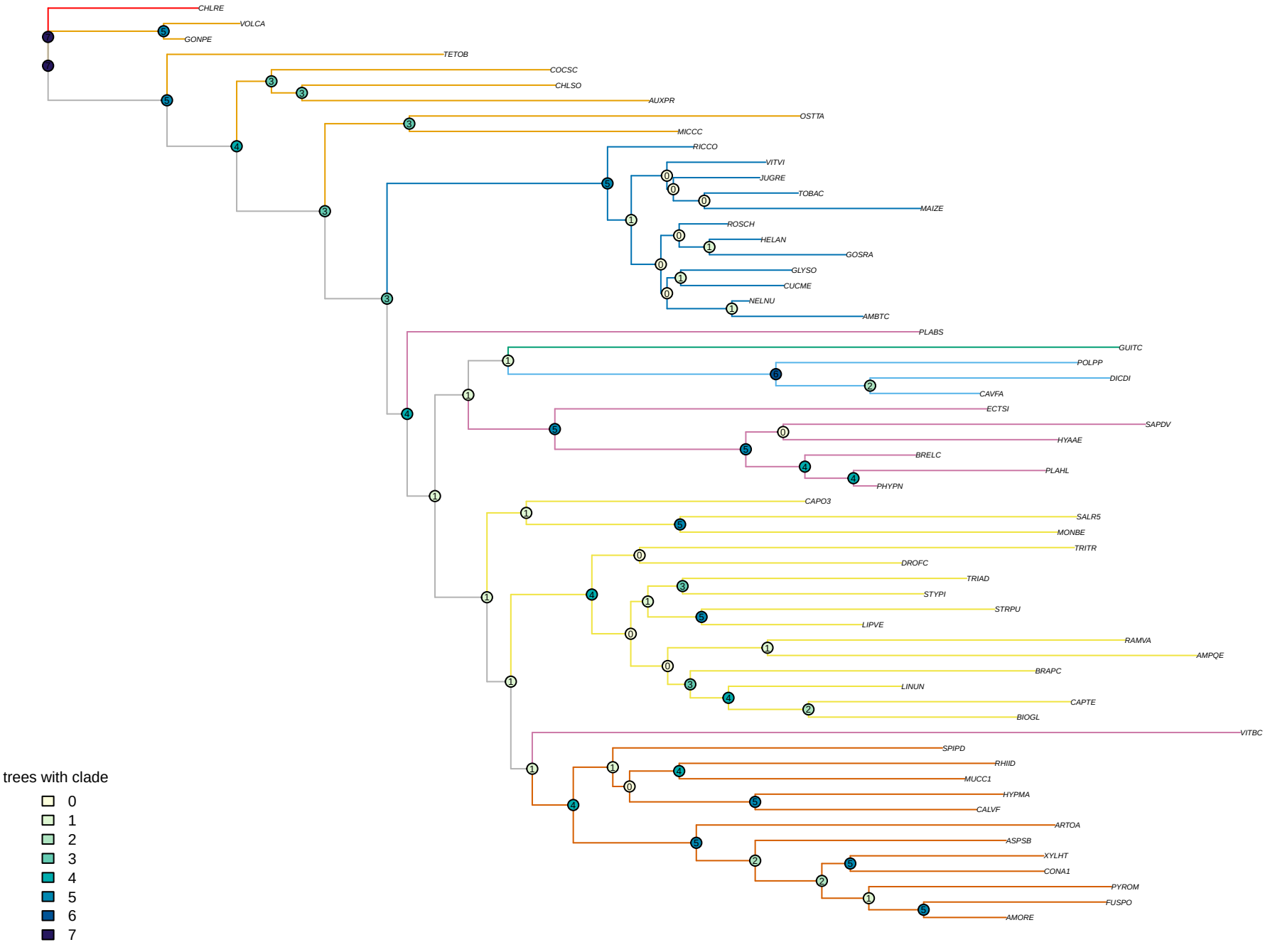

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
MSA−T embedding
Phyloformer
MSA−T attention
IQ−TREE
CMAPLE
EB−ncbi
EB−ncbi−astral
EB−astral
MSA−T embedding
Phyloformer
MSA−T attention
IQ−TREE
CMAPLE
EB−ncbi
EB−ncbi−astral
EB−astral


quartet consistency (8 trees; 29.5% shared by all)
share of quartets
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

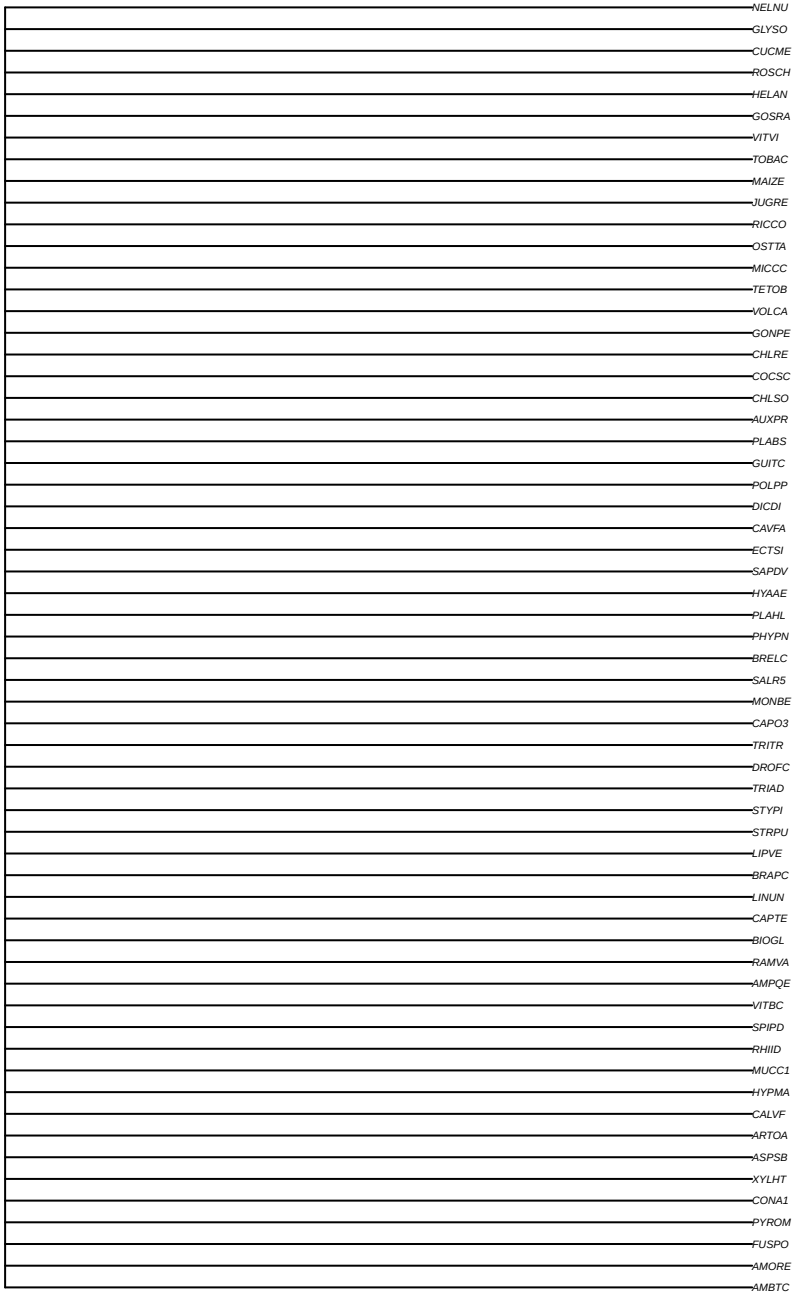

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
EB−astral
EB−ncbi−astral
EB−ncbi
CHLRE
VOLCA
GONPE
TETOB
COCSC
CHLSO
AUXPR
OSTTA
MICCC
RICCO
VITVI
JUGRE
TOBAC
MAIZE
ROSCH
HELAN
GOSRA
GLYSO
CUCME
NELNU
AMBTC
PLABS
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POLPP
DICDI
CAVFA
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SAPDV
HYAAE
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CAPO3
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MONBE
TRITR
DROFC
TRIAD
STYPI
STRPU
LIPVE
RAMVA
AMPQE
BRAPC
LINUN
CAPTE
BIOGL
VITBC
SPIPD
RHIID
MUCC1
HYPMA
CALVF
ARTOA
ASPSB
XYLHT
CONA1
PYROM
FUSPO
AMORE
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
CHLRE
VOLCA
GONPE
TETOB
COCSC
CHLSO
AUXPR
OSTTA
MICCC
RICCO
VITVI
JUGRE
TOBAC
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BIOGL
VITBC
SPIPD
RHIID
MUCC1
HYPMA
CALVF
ARTOA
ASPSB
XYLHT
CONA1
PYROM
FUSPO
AMORE
0
50
100
150
200
250
substitutions/site from root

IQ−TREE clades: number of the other 7 trees that contain each clade



strict



majority rule

